

The 3D Printing Success Guide

A practical, no-fluff guide to getting clean, reliable prints — for beginners and intermediates on any FDM printer.



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HOW TO USE THIS GUIDE

Skim the section titles first. When a print fails, jump straight to Section 4 — most problems have a one-line fix.

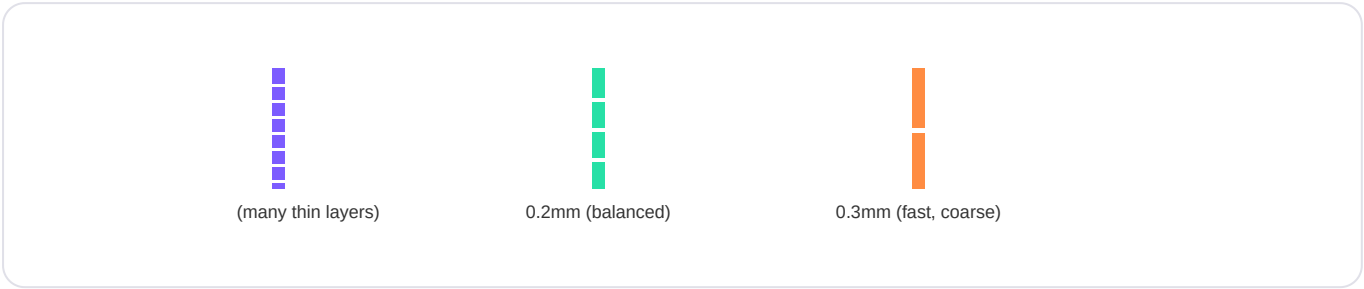
Slicer Basics: The Settings That Matter

Your slicer turns a 3D model into instructions your printer understands. These four settings affect quality more than anything else — and they work the same way regardless of which slicer or printer you use.

Layer Height

This is how thick each printed layer is. Lower = smoother surface, longer print time. Higher = faster, more visible lines.

Layer Height	Result	Best For
0.1mm	Very smooth, slow	Detailed miniatures, display pieces
0.2mm	Good balance	Most everyday prints (default choice)
0.3mm	Fast, visible lines	Large functional parts, prototypes



Infill Percentage

Infill is the internal lattice that fills the hollow interior of your print, giving it strength without using as much material as a fully solid part.

LIGHT-DUTY

10-15% infill is enough for decorative objects, figurines, and anything not load-bearing.

FUNCTIONAL PARTS

25-40% infill for brackets, hooks, or anything that needs to hold weight or stress.

Print Speed, Walls & Temperature

Print Speed

Faster isn't free — speed sacrifices detail and increases the risk of ringing (small wavy artifacts near sharp corners) and weak layer bonding.

- **40-60 mm/s** — safe general-purpose range on most machines
- **First layer: 20-25 mm/s** — slower always bonds better to the bed
- **Outer walls: ~50% of main speed** — this is the surface people actually see

If your printer ships with a tuned default profile, start there. Only raise speed after several prints come out clean.

Wall (Perimeter) Count

Walls are the solid outer shell of your print. More walls = stronger, more watertight, slightly slower.

- **2 walls** — default for most objects
- **3-4 walls** — functional parts, anything handled often, or anything holding liquid

Top/Bottom Layers

These are the solid layers that cap your infill at the top and bottom of the print. Too few and you'll see the infill pattern "ghosting" through the top surface — a common beginner complaint that has nothing to do with your printer and everything to do with this setting.

COMMON MISTAKE

Seeing small gaps or a waffle pattern on the top surface of a print? Increase top layers from 3 to 5-6 before touching anything else.

Print Temperature

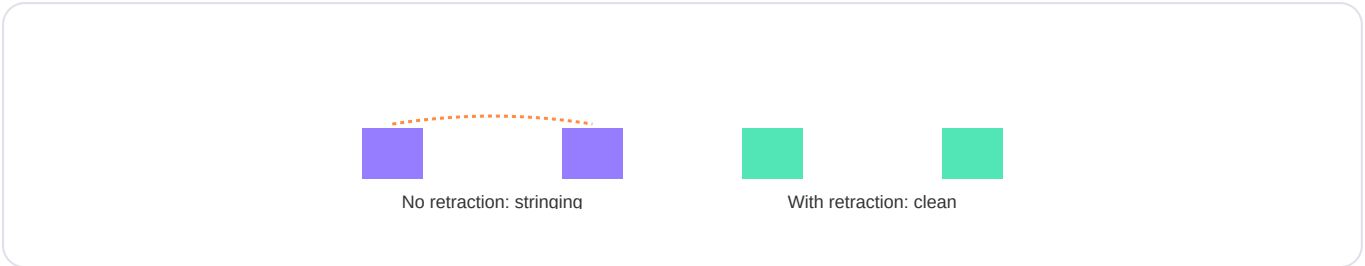
Set by filament type, not preference. Printing too cold causes weak layer bonding (parts snap easily); too hot causes stringing and blobbing. Start at the filament manufacturer's recommended temperature — it's usually printed on the spool itself — and adjust $\pm 5^{\circ}\text{C}$ only if you see specific issues covered in Section 4.

Retraction & Cooling

These two settings cause more confusion than any others — and they're the fix for several of the most common failures in Section 4.

Retraction

When the nozzle travels between separate parts of a layer, molten plastic keeps oozing out. Retraction pulls the filament back slightly during that travel so it doesn't leave strings behind.



Your correct retraction distance depends on which extruder type your printer uses:

Extruder type	How to tell	Retraction distance
Direct drive	Motor sits directly on the print head	0.5-2mm
Bowden	Motor on the frame, long tube to the head	4-7mm

DON'T OVERDO IT

Too much retraction causes clogs and filament grinding. If you increase it and prints get worse, go back down — more is not better.

Part Cooling Fan

The cooling fan freezes plastic in place as it's laid down. More cooling gives crisper detail and better overhangs, but weakens the bond between layers.

- **PLA:** 100% after the first layer
- **PETG:** 30-50% — too much cooling causes cracking and poor layer bonding
- **First layer, any material:** fan off, for maximum bed adhesion

QUICK SETUP CHECKLIST

Layer height 0.2mm → Infill 15-20% → Walls 2-3 → Top/bottom layers 4-5 → Speed 40-60 mm/s → Temperature per filament spool → Fan per material. This combination works for 90% of prints without further tuning.

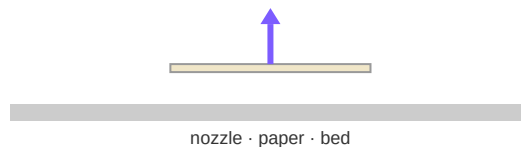
Bed Leveling & First-Layer Adhesion

Nearly every failed print traces back to one thing: a bad first layer. Get this right and most other problems disappear on their own.

Why the First Layer Matters

If the nozzle is too far from the bed, the plastic doesn't stick — the print detaches mid-way through. Too close, and the nozzle scrapes the bed, blocking flow or gouging your surface.

The Paper Test



1. Heat the bed and nozzle to printing temperature first (materials shrink/expand with heat)
2. Place a sheet of regular paper between the nozzle and bed at each corner + center
3. Adjust that corner's leveling knob until you feel slight friction pulling the paper through
4. Repeat all points — leveling is never a single-point adjustment

HEADS UP

Always level with the bed and nozzle already heated. A bed leveled cold will be measurably different once heat causes materials to expand.

IF YOUR PRINTER HAS AUTO BED LEVELING

Many newer printers use a sensor (BLTouch, CR-Touch, strain gauges) to auto-level and don't need the manual paper test. What you tune instead is the **Z-offset** — how close the nozzle sits to the bed after the sensor measures it. Symptoms are identical to manual leveling problems (see the diagnosis table on the next page): if the first layer doesn't stick, lower the Z-offset slightly (closer to bed); if it's scraping, raise it slightly.

Improving First-Layer Adhesion

Surface Prep

- **Clean the bed** — oils from fingerprints are the #1 hidden cause of poor adhesion. Wipe with isopropyl alcohol before every print.
- **Use a brim or skirt** — a brim adds a thin single-layer flange around your model's base, dramatically increasing surface contact for tricky, small-footprint prints.
- **Slow the first layer** — most slicers let you set a separate, slower speed just for layer one. Slower first layer = more time for plastic to bond to the bed.

Bed Surface Types

Surface	Cleaning	Notes
PEI (smooth or textured)	IPA wipe; wash with soap water occasionally	Most common on modern printers, good grip when clean
Glass	IPA or glass cleaner	Needs a glue stick or hairspray layer for PLA to stick well
Blue painter's tape	Replace when worn/dirty	Cheap, reliable fallback for any material

First-Layer Diagnosis

What you see	Likely cause	Fix
Plastic doesn't stick at all	Nozzle too far / dirty bed	Re-level, clean with IPA
Thin, see-through first layer	Nozzle too close	Raise nozzle slightly, re-level
Corners lift while printing	Warping (see Section 4)	Add brim, check enclosure/drafts
Rough, scraping sound	Nozzle too close	Stop print, raise nozzle

TIP

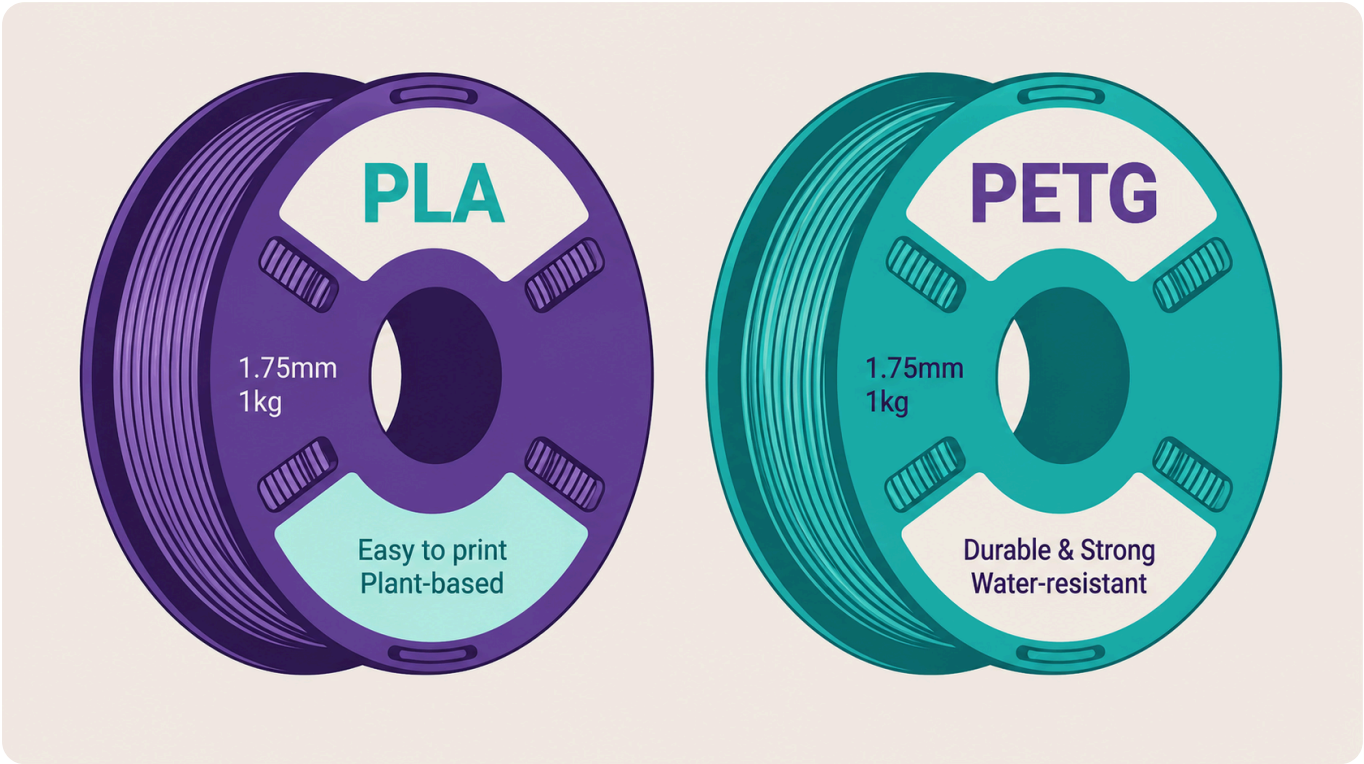
Watch the very first layer print in person at least once. Almost every future failure can be predicted — or prevented — right here.

Filament Basics

The Two Materials Beginners Should Know

Material	Ease	Strength	Best For
PLA	Easiest — low warp, no enclosure needed	Rigid but brittle under stress	Decorative prints, figures, most beginner projects
PETG	Slightly trickier — more stringing	Tougher, more flexible, more heat-resistant	Functional parts, outdoor items, anything under stress

If you're not sure which to use: start with PLA. It's the most forgiving material for learning your printer's behavior, and most of the files in this bundle print cleanly in PLA at default settings.



ALREADY HAVE ABS?

ABS is a common third material, but it's not beginner-friendly: it needs a heated enclosure (drafts cause severe warping), higher temps (~230-250°C nozzle, ~90-110°C bed), and ventilation — it releases fumes while printing. If you're new, print your first several files in PLA before touching ABS.

Filament Storage

THE HIDDEN FAILURE CAUSE

Filament absorbs moisture from the air. Wet filament causes popping sounds, stringing, weak layers, and rough surface texture — even when every other setting is correct.

- Store spools in a sealed container or bag with silica gel packs when not printing
- If a spool has been open for weeks and prints look suddenly worse, moisture is the most likely cause
- A cheap fix: a food-storage container with a couple of desiccant packs works fine as basic dry storage

Is Your Filament Actually Wet?

Don't guess — check:

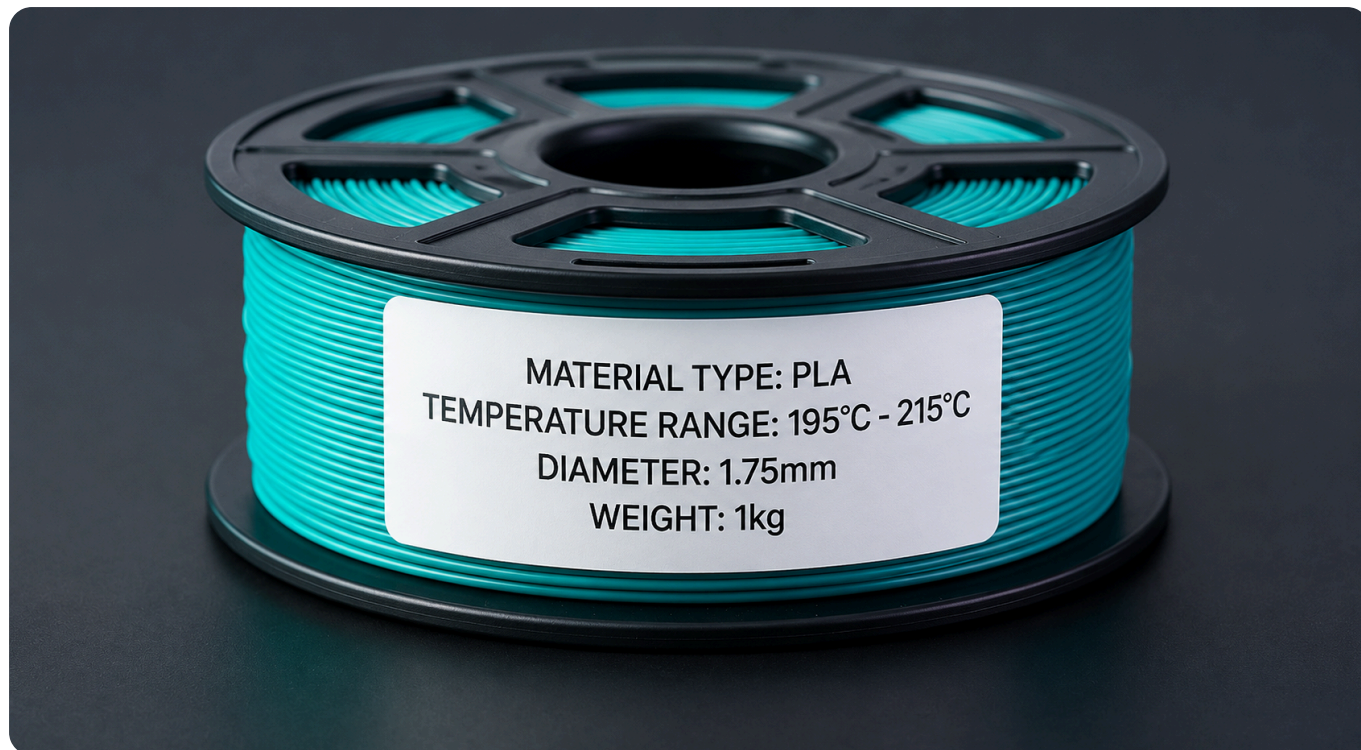
- **Listen while printing** — faint popping or crackling sounds near the nozzle mean trapped moisture turning to steam
- **Look at the surface** — wet filament often prints with a rough, bumpy, or slightly bubbly texture instead of smooth
- **Bend test** — dry PLA snaps cleanly; filament that's absorbed moisture tends to bend further before breaking, or breaks unevenly

DRYING WET FILAMENT

A filament dryer box is the reliable fix (typically 4-6 hours at ~45-55°C for PLA, higher for PETG — check your dryer's material presets). No dryer? A food dehydrator or an oven on its lowest setting with the door propped open works as a rough substitute — watch closely, filament can deform if it gets too hot.

Reading a Filament Spool

Every spool has a manufacturer-recommended range printed on it or in the packaging — this is your real starting point, not general internet advice.



PRINT TEMP

PLA: ~190-210°C. PETG: ~230-250°C. Spool ranges vary slightly by brand — start in the middle of the printed range.

BED TEMP

PLA: ~50-60°C. PETG: ~70-80°C. Too low = poor adhesion; too high = softening/warping.

Filament Diameter

Almost universally 1.75mm for hobbyist printers. Confirm your printer's spec before buying a new brand — the wrong diameter simply won't feed correctly, regardless of any other setting.

Color & Finish Notes

- Matte/silk finishes may need slightly different temperatures than standard finishes of the same material
- Darker colors sometimes absorb more heat during printing — if a specific color underperforms compared to others, try +5°C first

Common Print Failures & Fixes

Find what you're seeing below. Each fix is ordered — try the first bullet before the second.

Warping (corners lifting off the bed)



- Add a brim (5-8mm) in your slicer settings
- Clean the bed with isopropyl alcohol before printing
- Eliminate drafts — close windows, avoid fans blowing near the printer
- Increase bed temperature by 5°C

Stringing (thin hair-like strands between parts)



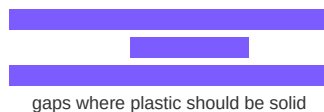
- Increase retraction distance: +0.5mm (direct drive) or +1-2mm (Bowden) from current setting
- Lower nozzle temperature by 5-10°C
- Dry your filament (see Section 3 — this is a very common hidden cause)
- Enable "combing" (keeps the nozzle inside the model when moving between points, reducing oozing) or "avoid crossing perimeters" in your slicer, if available

Layer Shifting (print suddenly offset mid-way)



- Check belt tension — a loose belt is the most common cause
- Reduce print speed, especially on outer walls
- Check that nothing (cables, filament) is snagging the moving parts
- Confirm pulley grub screws are tight on the motor shaft

Under-Extrusion (gaps, weak or missing layers)



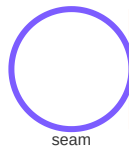
- Confirm filament diameter setting in slicer matches your actual filament (1.75mm)
- Check for a partial nozzle clog — see Section 7 for cleaning steps
- Increase temperature by 5°C — plastic may be too cold to flow properly
- Increase flow rate / extrusion multiplier by 2-5% (most slicers default to 100%)
- Confirm filament is feeding freely, not tangled on the spool

Over-Extrusion (rough surface, blobby dimensions, print slightly too "fat")



- Decrease flow rate / extrusion multiplier by 2-5% (test in 1-2% steps)
- Confirm nozzle diameter setting in slicer matches your actual nozzle (0.4mm is standard)
- Print a single-wall test cube and measure wall thickness with calipers to confirm actual vs. expected
- Lower temperature slightly if surface looks glossy/melted rather than crisp

Visible Z-Seam (a vertical line running up one side of the print)



This is where each layer's outer wall starts and stops — a small cosmetic seam, not a structural fault.

- In your slicer, set seam position to "aligned" or manually to the back of the model, so it lands somewhere less visible
- Enable "randomize" seam position if the seam matters less than a repeating pattern along one edge
- Enable coasting (stops extrusion just before reaching the seam point) if your slicer supports it — reduces the small blob that often forms there

Elephant's Foot (bulging first few layers)



base bulges wider than the rest of the print

- Lower bed temperature by 5°C for the first few layers only, if your slicer supports it
- Reduce first-layer horizontal expansion setting to -0.1 to -0.2mm
- Confirm nozzle isn't too close to the bed (recheck leveling — see Section 2)

SECTION 04 (CONTINUED)

Ringings / Ghosting (wavy pattern near corners)

- Reduce print speed and acceleration settings
- Check that the printer frame is on a stable, non-wobbly surface
- Tighten belts — same root cause as layer shifting, different severity

Gaps in Top Surface (infill pattern visible)

- Increase top layer count (see Section 1)
- Increase infill percentage slightly
- Check for under-extrusion — see above

Nozzle Clog / No Filament Coming Out

- Perform a "cold pull" to clear the clog — see Section 7 for the exact technique
- Check filament isn't wet or brittle (breaking inside the feed tube)

GENERAL RULE

Change one setting at a time between test prints. Changing three things at once means you'll never know which one actually fixed it.

Blobs & Zits (small bumps on the surface)

- Enable "combing" mode in slicer if available (avoids crossing open spaces unnecessarily)
- Reduce retraction distance slightly if it was set too aggressively
- Lower printing temperature by 5°C to reduce ooze

Print Detaches Mid-Print (not warping — comes loose entirely)

- Add or enlarge brim for more bed contact area
- Recheck bed leveling and cleanliness (see Section 2)
- Reduce initial layer speed for stronger first-layer bonding

Rough / Fuzzy Surface Texture

- Dry your filament — this is the single most common cause
- Reduce print speed on outer walls
- Slightly lower temperature if surface looks "melted" rather than crisp

Cracking or Splitting Along Layer Lines

- Increase nozzle temperature slightly for better layer bonding
- Reduce cooling fan speed — too much cooling weakens layer adhesion
- Increase wall count for added structural strength (see Section 1)

Print Comes Out Smaller / Larger Than Expected

- Confirm slicer scale is set to 100% (easy to accidentally resize a model)
- Check for consistent shrinkage across all axes — measure a calibration cube if unsure
- Confirm correct units (mm vs inches) when the model was originally designed

WHEN TO STOP AND RESTART

If the first 3-5 layers already look wrong, don't wait to see if it "fixes itself" — it won't. Stop, fix the setting, and reprint. You'll save more filament and time than letting a bad print run to completion.

Supports & Part Orientation

When You Need Supports

As a rule of thumb: any overhang steeper than roughly 45° from vertical will need support material underneath it, or it will sag or fail mid-print.

Fine, no support



Borderline (~45°)



Needs support



Support Settings That Matter

- **Support density** — 10-15% is usually enough; more just wastes material and time
- **Support interface layers** — a thin, denser layer directly touching your model, making removal cleaner and the surface smoother once removed
- **Support overhang angle threshold** — the angle at which your slicer decides to add support automatically; 45° is a safe default
- **Support Z-distance** — the tiny air gap between support and model, usually 0.1-0.2mm. Too small and supports fuse to your model, becoming very hard to remove; too large and the model prints without proper support underneath.

Support Type & Placement

Support Type: Grid vs. Tree



GRID / LINEAR (OLDER DEFAULT)

Straight vertical columns in a grid pattern. Simple and reliable, but uses more material and can be harder to remove from tight or curved areas.

TREE / ORGANIC (MODERN SLICERS)

Branches that grow toward overhangs like a tree, touching the model at far fewer points. Uses noticeably less filament, prints faster, and usually removes more cleanly — use this by default if your slicer offers it (Cura, PrusaSlicer, Bambu Studio, and most modern slicers do).

Support Placement

- **Everywhere** — supports overhangs anywhere on the model, including ones floating above other parts of the print. Use this for complex geometry.
- **Touching buildplate only** — only supports overhangs that have a clear path down to the bed. Use this when a model has internal cavities you don't want filled with support you can't reach afterward.

Choosing Part Orientation

How you orient a model on the bed affects strength, support needs, and surface finish more than almost any other decision you'll make.

Weak: layers stacked across the stress



Strong: layers align with the stress



orange arrow = direction of load/stress on the part

- **Minimize overhangs** — rotate the model to reduce steep angles needing support, when possible without sacrificing the design's function
- **Print strength runs along layers, not across them** — orient load-bearing parts so stress runs along the length of layers, not perpendicular to them, since layers separate more easily than they bend
- **Best surface faces up or forward** — supported/overhang surfaces are always rougher once support is removed; face the visible/important surface away from where support will touch

SIMPLE RULE

If a model has a "obvious" flat bottom, that's usually your best print orientation — flat surfaces need no support and adhere best to the bed.

Post-Processing

Removing Supports



- Use flush cutters or pliers to snap support material away from the main print, working slowly near detailed areas
- Let the print fully cool before removing supports — warm plastic is more likely to deform or tear
- Use a hobby knife to clean up any small nubs left after support removal

SAFETY NOTE

Always cut and pry away from your hands and fingers, not toward them — support material can snap suddenly once it gives way. Keep hobby knife blades sharp (a dull blade needs more force and is more likely to slip).

Sanding & Smoothing

Work through grits progressively — skipping straight to fine grit won't remove layer lines efficiently.

Grit	Purpose
150-220	Knock down visible layer lines
400	Smooth out scratches from the coarser pass
600-800 (optional)	Final polish before painting or clear coat

Always sand in the same direction as the layer lines first, then finish with light circular motions. Wet-sanding (dip paper in water) reduces dust and gives a slightly smoother result on the final pass.

Gluing & Finishing

Gluing Multi-Part Prints



BEST ADHESIVES BY MATERIAL

PLA: super glue (cyanoacrylate) works well. PETG: super glue or a PETG-compatible plastic cement. Always test on a scrap piece first, since some glues react differently to specific filament brands.

Painting (optional)

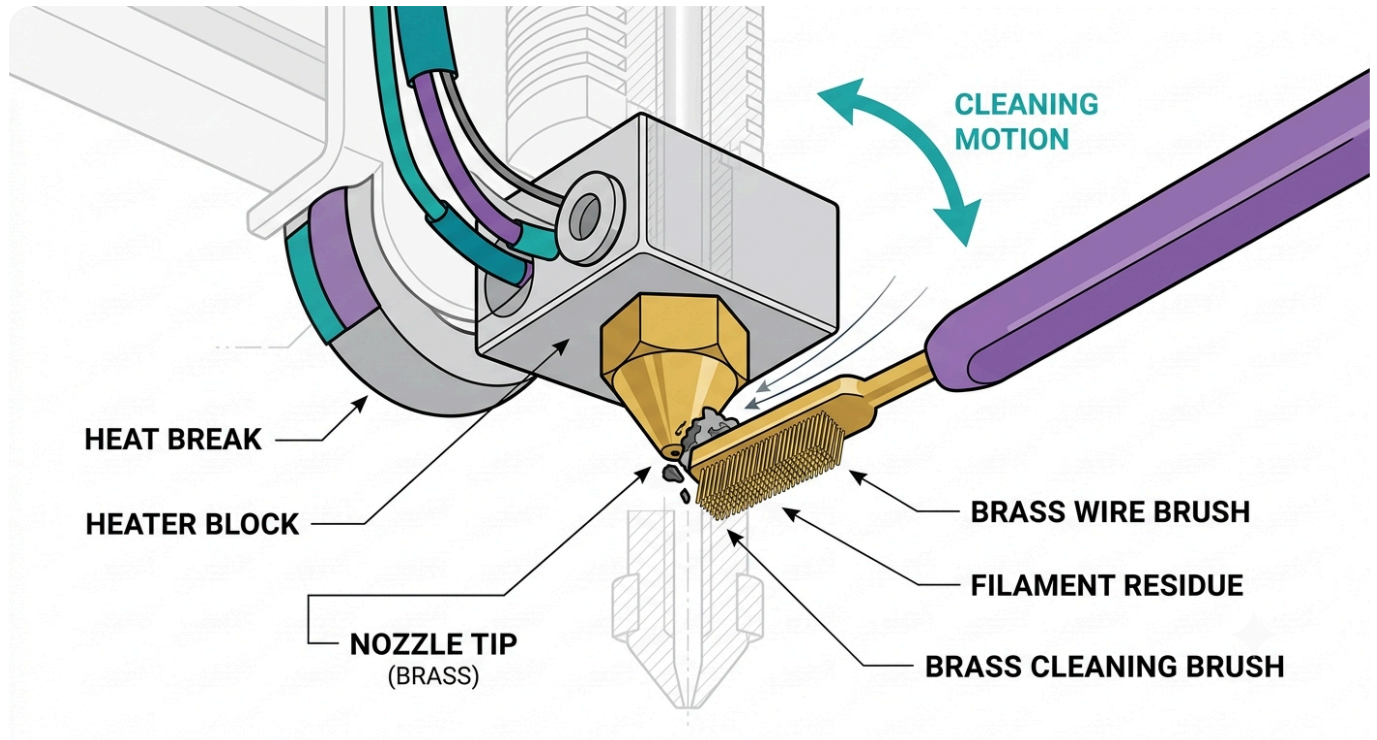
A light coat of filler primer evens out layer lines before painting and gives most acrylic or spray paints a surface to properly grip.

ORDER OF OPERATIONS

Remove supports → sand → glue multi-part assemblies → prime → paint. Doing these out of order (e.g., painting before sanding) means redoing work — sanding after paint removes the paint too.

Basic Maintenance

Weekly



- Wipe the build plate with isopropyl alcohol to remove oils and residue
- Check that the filament path is clear and feeding smoothly

Monthly

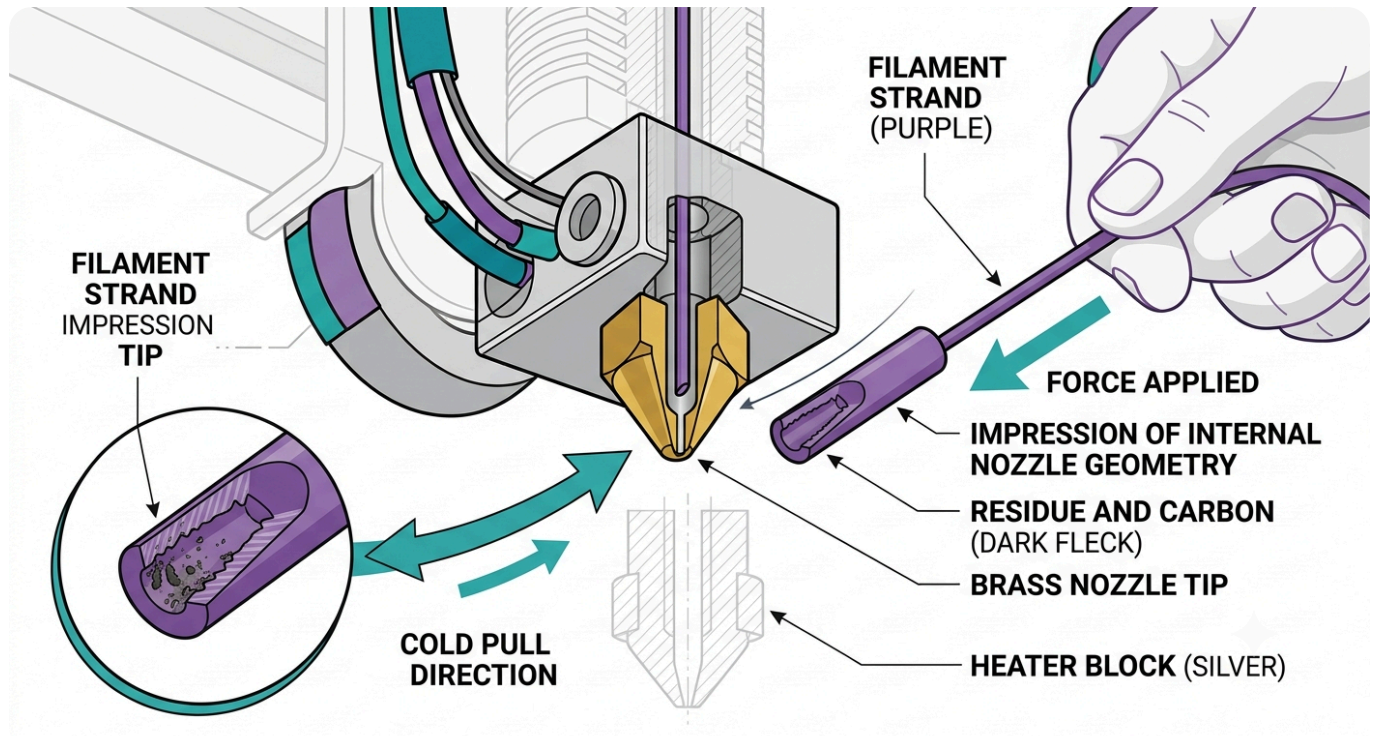


- **Check belt tension** — press the belt with a finger at its midpoint; it should deflect about 2-3mm, not feel loose/floppy and not be guitar-string tight. A quick pluck should give a dull thud, not a musical note.
- **Lubricate smooth rods/rails** if your printer's manual recommends it — use a PTFE-based or lithium grease made for linear motion, applied sparingly.
- **Clean the nozzle exterior** of any built-up residue

COMMON MISTAKE

Never use WD-40 or general household oil on rails or rods. It attracts dust and grit over time, which grinds into the rails and makes movement worse, not better.

As-Needed Maintenance



- **Nozzle replacement** — if extrusion becomes inconsistent even after cleaning, and you print often, nozzles wear out over time, especially with abrasive filaments
- **Cold pull** — the standard fix for a partially clogged nozzle: heat to printing temp, let cool ~10-20°C below that, then firmly pull the filament back out — repeat until it comes out clean
- **PTFE tube check (Bowden printers)** — if feeding becomes inconsistent, check that the tube is fully seated into its fittings at both ends; a gap here causes the same symptoms as under-extrusion (see Section 4) but no setting will fix it
- **Build surface wear** — if a previously reliable surface (PEI, textured sheet) starts losing grip despite cleaning, it's worn out and due for replacement — this is normal after months of regular use, not a sign you're doing something wrong

SAFETY NOTE

Always let the nozzle and bed cool before doing any hands-on maintenance near the hot end.

Quick-Reference Cheat Sheet

Setting	PLA	PETG	ABS
Nozzle Temp	190-210°C	230-250°C	230-250°C
Bed Temp	50-60°C	70-80°C	90-110°C
Layer Height (default)	0.2mm	0.2mm	0.2mm
Infill (general use)	15-20%	20-25%	20-25%
Print Speed	Default profile	Slightly slower than PLA	Slightly slower than PLA
Cooling Fan	100%	30-50%	0-20%
Enclosure	Not needed	Not needed	Required
Storage	Sealed w/ desiccant	Sealed w/ desiccant	Sealed w/ desiccant

Retraction Quick Reference

Extruder type	Retraction distance
Direct drive	0.5-2mm
Bowden	4-7mm

Fast Troubleshooting Index

Symptom	Go to
Corners lifting	Warping — Sec. 4
Hair-like strands	Stringing — Sec. 4
Sudden offset mid-print	Layer shifting — Sec. 4
Gaps / weak layers	Under-extrusion — Sec. 4
Rough surface, print slightly oversized	Over-extrusion — Sec. 4
Vertical line up one side	Z-seam — Sec. 4
Bulging base	Elephant's foot — Sec. 4
Wavy pattern near corners	Ringings/ghosting — Sec. 4
Waffle pattern on top surface	Gaps in top layers — Sec. 4
No filament coming out	Nozzle clog — Sec. 4
Small bumps on surface	Blobs & zits — Sec. 4
Print comes loose entirely	Print detaches — Sec. 4
Rough, fuzzy texture	Surface texture — Sec. 4
Cracking along layer lines	Layer bonding — Sec. 4
Wrong final size	Scale/shrinkage — Sec. 4
Won't stick to bed	Adhesion — Sec. 2
Supports won't remove cleanly	Support Z-distance — Sec. 5
Inconsistent feed (Bowden)	PTFE tube check — Sec. 7

KEEP THIS PAGE

Save or print this page separately — it's your fastest reference while you're standing at the printer.